

THE IMPACT OF ACCOUNTING SOFTWARE ON BUSINESS PERFORMANCE

Yvonne Chong

FTMS College Malaysia
yvonc80@gmail.com

Ismail Nizam, PhD

Senior Lecturer
FTMS College Malaysia
nizam@ftms.edu.my

Abstract

The objective of this paper is to investigate and explore the impact of Accounting Software on business performance of Malaysian firms. The study discusses and explores the effects of using AIS on the firm's performance. The result of this study is expected to help the firm's owners and manager in understanding the importance of using Accounting Information System (AIS) derived from Accounting Software to achieve performance. Several characteristics such as: efficiency, reliability, ease of use, data quality and accuracy influenced the use of AIS, thereby affecting the performance of firms. Modern literature and the result of this study shows that these AIS characteristics possessed by the accounting information such as: efficiency, reliability, ease of use, data quality and accuracy have significant effects on the use of AIS and firm's performance. Previous researches have shown that it is crucial for firms to use AIS to ensure the survival and sustainability of business in the increasingly competitive environment besides enhancing their business operations competency and efficiency. This study is one of few that could shed light on how the use of AIS affects the performance of firms. In this study, the authors propose that dimensions of using AIS are important for improve the performance of business organizations.

Keywords: Accounting Software, Accounting Information System, and Business Performance

1.0 INTRODUCTION

1.1 Background of the Research

Today, accounting has come to seize a significant position in its functions in the business world. Accounting plays a critical role in the operation of an organization. For every business, it is important that the financial information of the business activities is being kept up-to-date and monitored by the organization. Accounting involves various processes, ranging from simple to complex and even burdensome. Companies need to keep pace with constant changes in information technology in order to maintain a highly accurate and up-to-date accounting, statutory records and inventory (Igbaria, et al., 1997). Due to complexity of the accounting system increasing vulnerability to errors and the swelling volume of accounting transactions, a system that could process and store accounting data with increased speed, vast storage and processing capacity was necessitated. To satisfy the increase demand for up-to-date and accurate information, accounting software, an integration of accounting and information technology was introduced to the world (Wickramsainghe, et al., 2017).

Many past researches have shown that accounting information system adoption does increased firm's performance, profitability and operations efficiency in Malaysia, Finland, Spain, Iran and Pakistan (Saira et al., 2010; Gullkvist, 2002; Sajady et al., 2008; Kouser et al., 2011). New computer tools and information society enabled the companies in Turkey to effectively use their accounting system in dealing with customers and suppliers (Esmeray, 2016). Cragg et al. (2002) and Chan et al. (1997) found that alignment of IT strategy with business strategy increased firm's performance. In an earlier study, Raymond et al. (1995) found that firms perform better with alignment of organisational structure and IT structure.

There has no research been conducted in Malaysia to evaluate the impacts of using accounting software on the business performance of firms in Malaysia? Sam *et al.* (2012) did a research on the adoption of computerized accounting system in small medium enterprises in Melaka. However, there was no published research to explore the impacts of accounting software characteristic on the business performance.

1.2 Research Rationale

Accounting information systems (AIS) coupled by improved speed, accuracy and reliability and with good corporate governance practices lead to good firm performance (Ahmad, 2016; Hla and Teru, 2015). Literature has studied the firm performance and AIS alignment but few studies have looked at the relationship between characteristic of accounting software and firm performance. While the impact of corporate governance and strategic decision making on firm performance has been widely studied, the role of AIS in this relationship is remain unclear. Since AIS serves as the firm's business information infrastructure backbone, both transparency of corporate governance and strategic decision-making process depend on information obtained from the system. AIS with features error-free, user-friendly, easy to learn, technically sound, well documented and flexible are indirectly related to organization performance.

Accounting information system is of immense significance to both businesses and organization according to Hla and Teru (2015) as it facilitates management decision making, quality of the financial report, internal controls, the company's transaction besides playing a crucial role in economic system. The study asserts that businesses and organization should implement AIS because an efficient AIS yield sufficient accounting information which is critical for effective decision making and also ensures adequate, pertinent and true information are available to all levels of management to plan and control business activities. IS application impacted the organization in form of the benefits received. Through business performance, the impact of IT on organization is discerned (Brynjolfsson and Hitt, 1996 & 2000; Osei-Bryson and Ko, 2004) leading to business value (Lee, 2001).

An optimal implementation of AIS by firms means adapting more successfully to a changing environment and shows a high degree of competitiveness, thus enhancing the dynamic character of a company through assimilation use of AIS. In other words, there are improvements in administrative management regarding accountancy and finance. By using AIS, it is possible to gauge the risk of some operations or predict future earnings with sophisticated statistical software applications.

The purpose of this research paper is to analyse the influence of the accounting software characteristic on the firm's performance. The results of this paper will assist software developer to come up with new software that suit with user's need and also benefits the firms in acquiring the appropriate accounting software (Musa and Ahmad, 2004).

1.3 Research Aim and Objectives

The major purpose of this study is to assess the impact of using accounting software system on business performance of firms in Malaysia.

The specific objectives therefore are:

- (1) To investigate the impact of Software Efficiency on the Performance of Business*
- (2) To investigate the impact of Software Reliability on the Performance of Business*
- (3) To investigate the impact of Software Ease of Use on the Performance of Business*
- (4) To investigate the impact of Software Data Quality on the Performance of Business*
- (5) To investigate the impact of Software Accuracy on the Performance of Business*

1.4 Research Questions

In order to address the research problem, the following questions would be administered:

- (1) What is the impact of Software Efficiency on the Performance of Business?*
- (2) What is the impact of Software Reliability on the Performance of Business?*
- (3) What is the impact of Software Ease of Use on the Performance of Business?*
- (4) What is the impact of Software Data Quality on the Performance of Business?*
- (5) What is the impact of Software Accuracy on the Performance of Business?*

2.0 Conceptual Framework

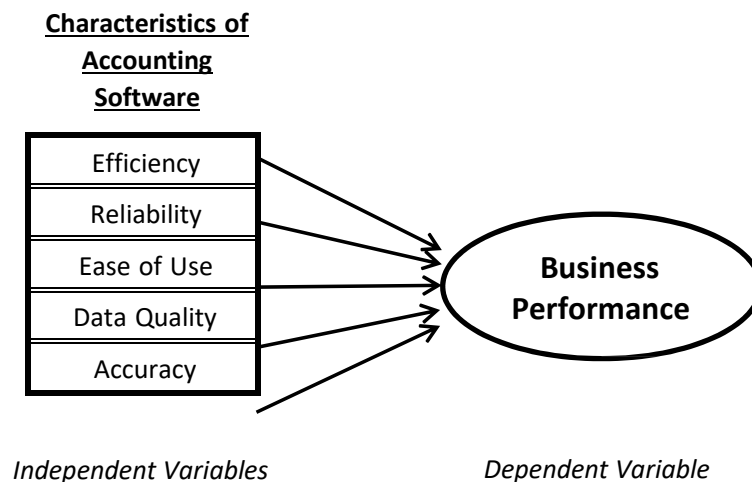


Figure 1: Conceptual Framework by the researcher

2.2.1 Efficiency and Business Performance

Accounting information system as a vital organizational mechanism is critical for effectiveness of decision management and control in organizations. An effective accounting information system as noted by Wilkinson et al., (2000) performs several key functions such as data collection and maintenance, accounting systems and knowledge management, data control and information generation. There are many factors that affect the efficiency and effectiveness of accounting information systems. Thus, the accounting information systems combined the factors qualified human resources, best software and hardware and data base quality to be effective.

Every business has pre-dominant goals to improve on performance and to maximize shareholder wealth. In order to achieve these objectives, right plans together with necessary resources are needed and used for implementation. Efficiency becomes very important considering the fact that resources are scarce. Efficiency in business context refers to ability of firm to maximize firm value by using the least inputs to achieve higher outputs. Empirical studies have reinforced the need for efficiency in the operations of the firm. Similarly, Greene and Segal

(2004) show that efficiency increase profitability of the firms in terms of return on equity in the insurance industry. There is a strong positive link between the firm's value and the utilisation of the firm's resources.

Therefore, we formulate the following hypothesis (see also Figure 1);

H1: There is a significant positive impact of Software Efficiency on Business Performance

2.2.2 Reliability and Business Performance

Accounting software produced reliability data that are critically used to plan, identify, and control business operations. As an essential characteristic for accounting information, reliability represents the extent to which the information is unbiased, free from error, and representationally faithful making it useful for decision making. Reliability is a complex and elusive construct of accounting information despite its central role (Maines and Wahlen, 2006). To achieve accounting standards, firms are required to provide more unabridged revelations associated to the underlying economic constructs represented by accounting information to help users' better determine the reliability of accounting information.

Reliability is ingrained in the information itself, and not in the manipulation of the information. The relevance of measurement attributes and economic constructs depicted by accounting information is a prerequisite for reliability to matter. Hence, reliability is an imperative but inadequate for accounting information to be functional. Reliability of accounting information determined by the requirement of accounting standards and facilitates firms to render economic constructs with pertinent informative accounting measurements and classifications. Thus, the usefulness of accounting information in predicting future cash flows depends on a number of factors such as accounting information reliability, the extent to which accounting constructs and measured values depict economic constructs without error or bias (Cho, Kim and Lim, 2006).

Chandar and Bricker (2002) in closed-end mutual funds predict and find that returns to market-wide portfolios S&P 500 Index and the Russell 2000 Index provide useful information to enable users to gauge the reliability of these funds' estimates of fair value gains and losses. Similarly, prior studies by Sunder and Waymire (1983), Casler and Hall (1985) and Shriver (1986, 1987) test reliability across measurement attributes under SFAS 33 using different industry-wide and economy-wide price-level indexes and baskets of assets. Various measures of historic stock return volatility used by Alford and Boatsman (1995) in testing the reliability of estimates of expected future return volatility to estimate fair values of firms' stock option-based compensation. The differences in these expected return volatility estimates triggering differences in the degree of reliability in estimates of options-based compensation expense. These studies imply that by revealing independent and verifiable benchmark data for underlying economic constructs make reliability more transparent and enhanced, thus enable financial statement users to gauge the reliability of reported accounting estimates for stock options, fair values, pension and other post-employment benefit obligations, loss reserves, and others.

Therefore, we formulate the following hypothesis (see also Figure 1);

H2: There is a significant positive impact of Software Reliability on Business Performance

2.2.3 Ease of Use and Business Performance

The quality of software which is ease of use and understandable, would be beneficial to its user. Therefore, the success of the system used depends on the

level of ease-of-use of the system. In general, an increase in ease of use positively influence several aspects of a company's output quality such as increased sales and revenues , productivity and customer satisfaction; reduced training and support cost, development time and costs and maintenance costs (Bias and Mayhew, 2005).

Landauer (1996) asserts that increasing ease of use in the workplace raises several responses from employees such as "Workers who enjoy their work do it better, stay longer in the face of temptation, and contribute ideas and enthusiasm to the evolution of enhanced productivity". Companies create standards by implementing experimental design techniques that create baseline levels. In an office environment, concerned areas include working posture, design of workstations, office environment and organization issues, input devices, screen displays, and software interface (Celine, 2008). By improving the said factors, firms can attain their goals with increased output at lower costs besides potentially produce optimum customer satisfaction level. There are numerous bases for correlation of these factors to overall improvement. For instance, easy to understand user interfaces software reduces the requirement for extensive training, improved interface tends to reduce the time needed to accomplish tasks, increase the productivity levels for employees and reduces development time and costs. Each factor is not mutually exclusive but rather they should be working in conjunction to form the overall workplace environment.

Usability classifies by Gould (1988) relating to the system performance, system functions and user interface (Dubey et al, 2010). System quality do not directly impact toward organisation performance but if system is technically sound, error-free, easy to learn, user-friendly, well documented and flexible those features are indirectly related to organization performance.

Therefore, we formulate the following hypothesis (see also Figure 1);

H3: There is a significant positive impact of Software Ease of Use on Business Performance

2.2.4 Data Quality and Business Performance

Output of accounting information system (AIS) very much depends on the data quality as poor data quality will result in garbage in garbage out (Xu, 2003). Through evaluation of accounting information quality using four attribute namely accuracy, timeliness, completeness, and consistency, Xu (2003) concluded that data quality is critical AIS, thus influencing the firm's performance. To achieve high data quality, the process of data production such as data collection, data utilization, and data storage must work satisfactorily (Lee and Strong 2003). According to (Xu, 2009), the competitiveness of firms would be damaged by incomplete and inaccurate data as input control and employee competencies are important to data quality of accounting information system. Huang, Lee, and Wang (1999) found that data of inferior quality may cause adverse effect on decision making and consequently affect firm's performance.

Several researches focus on success factors and critical process factors that frequently influence the information systems as well as AIS. Wixom & Watson (2001), data quality as one of the critical factor is relevant directly to effort for

decision making. In this information era, proactive approach to data quality management is required to enhance reliance of the organization on (AIS) to achieve their mission (Al-Hakim, 2007). Information and data are described as critical components for all the activities of every human venture (Emeka-Nwokeji, 2012). Literature has been made for determining and handling critical success factors at data quality management level as noticed by Xu (2003). Nevertheless, there have been few attempts to identify the critical data quality measures in the context of AIS causing data quality in AIS to remain largely unexplored. Emeka-Nwokeji (2012) concluded that data quality is significant for the success of AIS as the quality of data is assured for enhancing firm's performance. Data quality management policies aid companies in proactively responding and supplying services and products required by the customers, and appropriate processes of decision and operation. Few studies have scrutinized the effect of the AIS' data quality as critical success factor on organisational performance. Al_Qudah and Shukeri (2014) noted data quality is strongly related and positively impacts the perception of company internal auditors. More scientific studies on found data quality and AIS performance are strongly related (Emeka-Nwokeji, 2012).

Therefore, we formulate the following hypothesis (see also Figure 1);

H4: *There is a significant positive impact of Software Data Quality on Business*

Performance

2.2.5 Accuracy and business performance

Accounting software is not only speedy but also accurate and reliable. Computers are being used to collect data and process into meaningful information that management use to make timely and effective decisions and the entire data processing are carried out using accounting software through sorting, classifying calculating, summarizing the data and production of reports.

Information systems processing and production processing in manufacturing organizations are homogeneous. Firm will lose business when customers (users) will be dissatisfied if the product (information) is not delivered on time (timeliness) and the product (information) does not conform to the needs (relevance) of customers (users) (Clikeman, 1999). Non conformation of information provided by an IS leads to disruption of operations in the organization and hefty maintenance costs, ensuing in high costs to the organization. When the operational information from accounting software is highly accurate, firm will benefit from reduced labour costs and waste, effective use of machinery, and low inventory costs. Thus, information content with high accuracy, completeness and relevancy leads to better product cost control and increase organizational efficiency through increased profit margin and decision making efficiency.

Ravichandran and Rai (2000) assert by leveraging IT correctly, internal operational efficiencies of organization can be achieve through efficiently and effectively manage of internal resources and improve customer service to attain strategic advantages. Thus, accuracy of accounting software is highly important to organizations in terms of benefit through value derived by providing accurate and timely information to improved decision making, identifying possible future business avenues and profitable ventures to. Dependency on information systems

and losses associated with accuracy of information urge management to improving IS quality deliberately (Ravichandran and Rai, 2000).

The information is accurate and credible when it does not contain significant errors, it is not biased, and users can trust that it accurately represents what it has set out to be or what they expected to. The information must represent accuracy of the transactions and other events that are intended to be represented and is reasonably expected to represent them. With the aid of the software, accountants tend to improve the overall accuracy of their record thus eliminating or reducing human error. Accounting involves a lot more of math calculations done by hand. A little mathematical error in the computation at the beginning of this process can significantly affect the result. But accountants can process data accurately using accounting software to provide complete, accurate and timely information outputs for decision making in driving business efficiency and growth. Accuracy of financial data is consistency and efficiency driver across the entire organization enhancing the company's performance and the achievement of key business goals, operationally and financially.

Therefore, we formulate the following hypothesis (see also Figure 1);

***H5:** There is a significant positive impact of Software Accuracy on Business Performance*

3.0 RESEARCH METHODOLOGY

This section discussed about the research design and methodology employed to conduct this research.

3.1 Research Paradigm

This study was guided by the positivism paradigm. Positivism is based on values of reason and validity and focuses on facts garnered from theories and which can be measured empirically using quantitative methods. The positivist research approach assumes that only knowledge that is based on observations of the external reality is valid (Phillips & Burbules, 2000). Positivism also assumes that theories can explicate the relationship between dependent and independent variables and developed hypothesis to be tested and forecast the outcomes of the study. Therefore, the impacts of accounting software characteristics had on the business performance in Malaysia firms were able to explore by using positivism paradigm.

3.2 Research Design

Explanatory research is concerned with determining the causal relationship between variables (Zikmund & Babin, 2010).

The purpose of this research is more on seek new insights and determining the causal relationship between variables based on exploratory and explanatory. It is exploratory because researcher is exploring the relationship of independent and dependent variables in an unstructured and informal manner collection of information and explanatory because researcher explained the impact of dependent and independent variables (Hair, Rabin, Money and Samouel, 2003).

Hence, the research was conducted using combination of exploratory and explanatory research design.

3.3 Research Method

In this study, quantitative research approach was adopted as it allows collecting more data to investigate the facts, testing theories and hypotheses. As a result, quantitative is more suitable to represent tested theories comparing qualitative method since researcher was unaware of essential variables to analyse and the sensitivity of data collection relating to the social settings. Using this method allows researcher to streamline research problem to restricted variables and supports to understand the result of analyst (Creswell, 2003). This type of research is more objective and reliable as the researcher cannot interpret the facts of the research based on biased opinion.

3.4 Research Instrument

The research used close – ended questions in order to source information from responses. The questionnaire was structured in a five point likert format to extract the data or information.

A questionnaire is a set of questions which are usually sent to the selected respondents to answer at their own convenient time and return back the filled questionnaire to the researcher. In this study, questionnaires were used to collect information from respondents. The reason for using questionnaires is because they cover large sample at low cost, and gives respondents adequate time to give well thought-out answers.

3.4.1 Data Collection Method

A total 150 questionnaires were distributed, 100 returned and out of 100, 22 questionnaires were rejected due suspicious answers. This research used cross-sectional data collection technique which is a form of quantitative sampling methods used in explanatory research (Brady and Johnson, 2008).

The sampling technique used in this research was random probability sampling method which is popular among researchers as this technique eliminates prejudices by allowing the results to exceed depending on sampling population. On the contrary, respondents are selected by researcher in non-probability sampling method may have impact on validity of the results.

3.5 Data Analysis Plan

Data analysis is very crucial path of any research involving process of inspecting, cleansing, transforming, interpreting and modeling data with the goal of discovering useful information to support the hypothesis underlying each question (Judd & McClelland, 1989) depending on nature of the sampling, measurement and data collecting method. Quantitative statistical analysis for questionnaire was performed using Statistical Package for Social Sciences (SPSS) (IBM Corp, 2015) due to its effectiveness in managing data with wide range of options, generate better results and is specifically designed to analyse statistical data (Green and Salkind, 2010).

In determining the objective of this research, statistical procedures are conducted using Statistical package for social science (SPSS) to measure accuracy of the data.

It helps to analyse data more effectively and efficiently. Five statistical methods namely Descriptive Analysis, Normality Analysis, Reliability Analysis, Exploratory Factor Analysis and Multiple Regression were employed to examine data for this study.

4.0 ANALYSIS RESULTS AND FINDINGS

Section four presents data analysis, interpretation of results and hypotheses testing for this study.

4.1 Demographic Analysis

The table below showed the distribution of demographics for the respondents. 37.2% respondents were males and 62.8% respondents were female. The collected data revealed that more female workforce is using accounting software at their workplace. A total of 52 respondents have been using Accounting software for more than 7 years. The demographic profile also showed that all firms (100%) adopted computerized system for their business and a significant number 98.7% used computerized system to record data. Total of twenty-seven (27) respondents were working with the same firm for 5 years, fifteen (15) respondents below 15 years and eighteen (18) respondents each were working below ten (10) years and more than fifteen years respectively. In Malaysia, the most adopted software package was SAP with twenty-one (21) firms using it for their business, followed by Autocount (18) and Sage (14) while the least adopted software package was Peachtree with only three (3) firms using it according to this study.

Table 1: Demographics of the study

Variables	Measures	Frequency	Percent	Valid Percent	Cumulative Percent
Gender	Male	29	35.8	37.2	37.2
	Female	49	60.5	62.8	100.0
Experience in Using Accounting Software	0 - 2	5	6.2	6.4	6.4
	3 - 4	7	8.6	9.0	15.4
	5 - 6	14	17.3	17.9	33.3
	Over 7	52	64.2	66.7	100.0
Years of Service in current firm	Up to 5	27	33.3	34.6	34.6
	6 – 10	18	22.2	23.1	57.7
	11 – 15	15	18.5	19.2	76.9
	Over 15	18	22.2	23.1	100.0
Firm used Computerised System?	Yes	78	96.3	100.0	100.0
Firm used Computerised Data Recording?	Yes	77	95.1	98.7	98.7
	No	1	1.2	1.3	100.0
Software Package used by current firm	Sage	14	17.3	17.9	17.9
	Quick Books	5	6.2	6.4	24.4
	SAP	21	25.9	26.9	51.3
	Peachtree	3	3.7	3.8	55.1
	Autocount	18	22.2	23.1	78.2
	Others	17	21.0	21.8	100.0

4.2 Descriptive Analysis and Normality Analysis

This research comprise of five independent variables and one dependent variable. The dependent variable is business performance which is measured using efficiency, reliability, ease of use, data quality, and accuracy. The descriptive statistics analysis table below showed that R, DQ and EOU were found to affecting BP with the mean value of 4.39, 4.79 and 4.33 each and standard deviation of

0.631, 0.578 and 0.605. A and Ef mean value are 4.14 and 4.11 respectively with standard deviation 0.708 and 0.642 are also two of the important independent variables that least affects BP (See Table 2). All the means and standard deviation are ± 1 shows that the data set is good.

Table 2: Descriptive and Normality statistics generate from SPSS

	N	Minimum	Maximum	Mean	Std. Deviation	Skewness		Kurtosis	
	Statistic	Statistic	Statistic	Statistic	Statistic	Statistic	Std. Error	Statistic	Std. Error
Business Performance (BP)	78	3	5	4.05	.597	.015	.272	-.593	.538
Accuracy (A)	78	2	5	4.14	.708	-.821	.272	.091	.538
Reliability (R)	78	3	5	4.39	.631	-.791	.272	-.443	.538
Efficiency (Ef)	78	2	5	4.11	.642	-.662	.272	.183	.538
Ease of Use (EOU)	78	3	5	4.33	.605	-.610	.272	-.392	.538
Data Quality (DQ)	78	3	5	4.37	.578	-.792	.272	.236	.538
Valid N (listwise)	78								

The above table shows the five (5) independent variables (A, R, Ef, EOU, DQ) are negative values and the dependent variables BP show positive values. The independent variable A were highly skewed with negative skewness value -0.821 indicates a long left tail and a significant sharp-rising center peak compared to normal distribution and positive Kurtosis value of 0.91 indicates a leptokurtic distribution with a fatter tails. The independent DQ and Ef were also highly skewed with each skewness value -0.792 and -0.662 while Kurtosis value 0.236 and 0.183 indicates a long left tail and fatter tail similar to A variable. The independent variable R and EOU were also another highly skewed distribution with skewness value -0.791 and -0.610 while Kurtosis value -0.443 and -0.392 respectively indicates similar long left tail but a platykurtic distribution with thinner tail.

Dependent variable BP shown above were positively skew with the skewness value 0.015 and negative Kurtosis value -0.593 indicates a long right tail and the central peak is higher and sharper than the normal distribution (Leptokurtic). The skewness and Kurtosis value were within ± 1 indicates the distribution for this study is normal (Hair et al., 2010).

4.3 Reliability Analysis

Reliability is a test helps to ensure the overall consistency of measures. A general accepted rules describing internal consistency using Cronbach's alpha is as follows: Excellent ≥ 0.9 ; Good ≥ 0.8 ; Acceptable ≥ 0.7 ; Questionable ≥ 0.6 ; Poor ≥ 0.5 ; Unacceptable ≤ 0.5 which may vary depending on the sample size (DeVellis, 2012; Hair et al., 2010). Pilot study was conducted prior to the main study enhanced the likelihood of success of the main study and potentially help to avoid failure in main studies (Van Teijlingen and Hundley, 2001). The Table 3 shows all constructs in this research are reliable where the values are ranged between 0.80-0.90. Variables R and BP were excellently reliable to conduct the study with Cronbach's alpha figure above 0.90. Similarly, reliability of Ef, EOU, DQ and A were in good state with Cronbach's alpha above 0.80 to conduct this research. All the value of Cronbach's alpha coefficients met with requirements of reliability to proceed with further analysis.

Table 3: Cronbach's alpha coefficients (SPSS)

Constructs	Number of Items	Cronbach's Alpha
All Variables	42	0.975
Efficiency	7	0.879
Reliability	9	0.924
Ease of Use	6	0.821
Data Quality	6	0.857
Accuracy	7	0.858
Business Performance	7	0.911

4.4 Exploratory Factor Analysis (EFA)

Exploratory factor analysis was conducted after confirmed that the data was suitable for factor analysis, with principal components analysis (PCA) as the method of factor extraction, to identify the underlying factor structure of the accounting software characteristic and business performance. The PCA summarizes the interrelationships among the original variables in smaller set of underlying dimensions. As in this case it is used when the researcher has no pre-existing knowledge about the factors that may explain the correlations between a set of variables.

4.4.1 Correlation Matrix and Anti-Image Matrix

The original 42 variables were subjected to principal component analysis and orthogonal varimax rotation with Kaiser Normalization. The suitability of data for factor analysis was assessed prior to performing the EFA. Inspection of the correlation matrix revealed the presence of many coefficients of 0.3 and above. The correlation matrix produced a significant number of large correlations suggesting that factor analysis is a relevant statistical methodology. Moreover, each of the diagonals of the anti-image correlation matrix was above 0.7.

4.4.2 Sample Adequacy Analysis

The Kaiser–Meyer–Olkin (KMO) test and Bartlett test of Sphericity were undertaken. These tests are used to establish the adequacy of the item correlation matrix upon which factor analysis is based. Data were analysed using the IBM SPSS Statistics V21.

The KMO coefficient (Table 4) for this dataset fell at Meritorious Level 0.855 (in between 0.80 to 0.89) exceeding the recommended value of 0.6 (Kaiser 1974), indicating that the sample is adequate (Hutcheson & Sofroniou, 1999). The approximate of Chi-square is 3307.151 with 741 degrees of freedom and Barlett's Test of Sphericity is significant at 0.000 which is less than 0.05 (Cerny & Kaiser, 1977) indicating that properties of the correlation matrix justified factor analysis to be used. Hence Factor Analysis is considered as an appropriate technique and valid for further analysis of the data.

Table 4: KMO and Bartlett's Test (SPSS)

Kaiser-Meyer-Olkin Measure of Sampling Adequacy.		.855
Bartlett's Test of Sphericity	Approx. Chi-Square	3307.151
	df	741
	Sig.	0.000

4.4.3 Communalities

Principal Components Analysis was deployed to extract the communalities. The communality for a variable is the variance accounted for by all the extracted factors. The higher the communality indicating the variable is more reliable. The mean level of communality is preferable to be at least .70 and for communalities not to deviate over a wide range (MacCallum, Widaman, Zhang & Hong, 1999). The mean communality for the 42 variables in this study is .78.

4.4.4 Total Variance Explained, Scree Plot and Rotated Component

Matrix

The results indicated that the total variance explained by each of the extracted components. A component is represented by all the variation in each of the variables. Each variable is standardized with the maximum variance for each as 1.0. An eigenvalue reflects the proportion of variance explained by the component. Kaiser's Criterion (Kaiser, 1960) specifies that only components with an eigenvalue of 1.0 or greater should be retained for analysis. Kaiser's Criterion is the default retention method in SPSS. Conway and Huffcut (2003) found that among organizational researchers, Kaiser's Criterion was the most highly utilized method of identifying the number of components to use in conducting a factor analysis. Higher percentages of total variance explained is an indicator that a strong relationship exists among a group of variables under study.

Before conducting factor analyses, we consulted peer-reviewed literature on the effect of using accounting software (; Ali, Omar and Bakar, 2016; Ismail and King, 2005). We noted that 10 of the 42 variables were designed to function as validity checks to address a number of factors that shape the context for interpreting the impact accounting software characteristics. The initial rotated solution revealed 7 components with an eigenvalue of 1.0 or greater that explained 77.9% of the variance. Inclusion of the 10 validity check items in this analysis, which were dispersed across the 7 factors and one variables X7 did not contribute meaningfully to the factor structure, posed a likely confound to the solution, and was therefore removed. The remaining 32 variables were submitted to a second EFA using the same procedure, yielding a 7-factor solution that accounted for 78,4% of the variance. Despite the slight increase in variance explained, the number of factor remained same at 7 which vary from the theoretical factor of 6. Therefore, problem variables, X25 which loaded to component 7 and the cross loading variable, X37 were eliminated, leaving 39 variables. The loading factor of above 0.65 is required for significance for a sample of 78 respondents (Hair et al., 2010).

The same EFA procedure was repeated on the 39 retained items, resulting in a 6-factor solution that explained 76.6% of the overall variance. Examination of the scree plot suggested that a six-factor solution provided the best fit. The suitability of a six-factor solution was also evident from initial eigenvalues and total variance associated with the six individual factors. The final rotated factor matrix for the EFA is presented in Table 4. Results showed each item correlated strongest with the factor to which it was assigned.

The first Factor, which was labelled Business Performance, describes the degree of impact from the use of accounting software. This encompasses nine items covering issues such as support, competencies, capabilities, system benefit and user satisfaction. This factor accounted for 50.78% of the total variance and had an Eigenvalue of 19.8. The Cronbach's alpha coefficient for this factor was 0.924. The second Factor was labelled Accuracy and accounted for 8.8% of the total variance and had an Eigenvalue of 3.446. The Cronbach's alpha coefficient was 0.949. The third factor was labelled Reliability and accounted for 5.4% of the variance and had an Eigenvalue of 2.116. The Cronbach's alpha coefficient for this Factor was 0.915. The forth factor was labelled Efficiency and accounted for 4.8% of the variance and had an Eigenvalue of 1.888. The Cronbach's alpha coefficient for this Factor was 0.927. The fifth factor was labelled Ease of Use and accounted for 3.7% of the variance and had an Eigenvalue of 1.453. The Cronbach's alpha coefficient for this Factor was 0.89. The last factor was labelled Data Quality and accounted for 2.96% of the variance and had an Eigenvalue of 1.152. The Cronbach's alpha coefficient for this Factor was 0.816. The Cronbach's alpha coefficient for the whole Index was 0.973.

4.5 Regression Analysis

This research performed a multiple regression analysis using SPSS to examine the impact of accounting software on business performance based on the five independent variables of accounting software characteristics efficiency, reliability, ease of use, data quality and accuracy. The dependent factor in this study is business performance. Model summary produced by linear regression explains the significant of the overall research model in achieving the desired output (Hair et al., 2010).

4.5.1 Model Summary

Table 5: Regression Model Summary (SPSS)

Model	R	R Square	Adjusted R Square	Std. Error of the Estimate	Durbin-Watson
1	.843 ^a	.711	.691	.33183	1.546

a. Predictors: (Constant), Data_Quality, Efficiency, Accuracy, Ease_of_use, Reliability

b. Dependent Variable: Business_Performance

The Model Summary (Table 5) above predicts that R as 0.843, R Square showing impact of accounting software characteristics on business performance of firms in Malaysia as 0.711 and adjusted R Square 0.691, indicating that 69.1% of the variance of accounting software characteristics can be predicted by the independent variables of efficiency, reliability, ease of use, data quality and accuracy. According to Zygmunt & Smith (2014), in normal terms a healthy variation dependent variable must be at least 60%, thus this model is found to be a good fit as it predicted above 60% of the entire model. Lastly, the Durbin Watson is found to be 1.546 which clearly indicates no auto correlation amongst the chosen sample. The Durbin-Watson test for auto-correlation shows that the value of the model is at 1.546 which is between the values 1.5 to 2.5 indicates that no auto correlation amongst the chosen sample (Hair et al., 2010). This shows that each sample is separately independent and not being influenced by the other samples.

4.5.2 Model Significance

The ANOVA statistics is used to represent the regression model significance. As in Table 6, the significance value for the F statistics is 35.48 and the significance ratio is 0.000 which is less than 0.05, which concludes that the regression model is statistically significant (Hair et al., 2010).

Table 6: Model Significance (SPSS)

ANOVA ^a						
Model		Sum of Squares	df	Mean Square	F	Sig.
1	Regression	19.532	5	3.906	35.478	.000
	Residual	7.928	72	.110		
	Total	27.460	77			

a. Dependent Variable: Business Performance

b. Predictors: (Constant), DataQuality, Efficiency, Accuracy, Ease of use, Reliability

4.5.3 Hypothesis Testing

Table 7 displays the results of hypothesis testing for the five independent variables based on the significant value from regression analysis, conducted using SPSS tool. A coefficient summarizes the parameters of the regression model. The standardized beta coefficients show the impact of each independent variable on the dependent variable (Business Performance).

Table 7: Coefficient

Model	Unstandardized Coefficients		Standardized Coefficients	t	Sig.	Collinearity Statistics	
	B	Std. Error	Beta			Tolerance	VIF
1 (Constant)	.405	.319		1.270	.208		
Accuracy	.055	.075	.066	.739	.462	.509	1.965
Reliability	.112	.102	.118	1.101	.275	.348	2.875
Efficiency	.428	.096	.460	4.441	.000	.373	2.677
Ease of use	.361	.102	.365	3.547	.001	.378	2.643
Data Quality	-.090	.091	-.087	-.983	.329	.516	1.937

a. Dependent Variable: Business Performance

As presented above, software Efficiency and Ease of Use has a moderate and significant impact of 46% and 36.5% on Business Performance respectively with the p-values (Sig.) less than 0.05. Therefore, if software efficiency and ease of use increase, business performance will increase. However, software Data Quality has a negligible -8.7% impact on Business Performance which is also not statistically significant as p-value (Sig.) is more than 0.05. The standardized beta coefficient for software Accuracy and Reliability are 0.066 and 0.118 shows a 6.6% and 11.8% of a positive impact of the independent variable (Accuracy and Reliability) on dependent variable (Business Performance) with a significance value of 0.462 and 0.275 which is more than 0.05. This indicates that Accuracy and Reliability places an insignificant impact on business performance of firms in Malaysia.

The following Table 8 shows the summary of Hypotheses Testing Results.

Table 4: Hypotheses Testing Results

Hypotheses	Beta Coefficient	Significant (P<0.05)	Decision	Interpretations
H1: There is a significant positive impact of Software Efficiency on Business Performance	0.46	0 Significant as the calculated p-value is less than 0.05.	Accepted	The beta coefficient of 0.460 indicates that Software Efficiency has a 46.0% positive impact on business performance.
H2: There is a significant positive impact of Software Reliability on Business Performance	0.118	0.275 Not Significant as the calculated p-value is more than 0.05.	Rejected	The beta coefficient of 0.118 indicates that Software Reliability has a 11.8% positive impact on business performance.
H3: There is a significant positive impact of Software Ease of Use on Business Performance	0.365	0.001 Significant as the calculated p-value is less than 0.05.	Accepted	The beta coefficient of 0.365 indicates that Software Ease of Use has a 36.5% positive impact on business performance.
H4: There is a significant positive impact of Software Data Quality on Business Performance	-0.087	0.329 Not Significant as the calculated p-value is more than 0.05.	Rejected	The beta coefficient of 0.329 indicates that Software Data Quality has a 32.9% positive impact on business performance.
H5: There is a significant positive impact of Software Accuracy on Business Performance	0.066	0.462 Not Significant as the calculated p-value is more than 0.05.	Rejected	The beta coefficient of 0.066 indicates that Software Accuracy has a 6.6% positive impact on business performance.

4.6 Discussions

4.6.1 Efficiency (Ef)

The standardized beta coefficient is 0.46 which shows a 46% positive impact of the independent variable (Efficiency) on dependent variable (Business Performance) with a significance value of 0.00 which is less than 0.05. This indicates that software efficiency places a significant impact on the business performance of business firms in Malaysia.

To ensure the relevancy of this result further, researcher have deep dive into other similar research to examine how those research could further support this research. According to Information Processing Theory utilized by Rogers et al. (1999), describe business organizational behaviour and present a model describing how effective and ineffective business strategies are developed. Levy et al. (2011) stated that firms have highest alignment and good performance with improves system efficiency. Study on IS implementation in India conducted by Sharma and Bhagwat (2003) suggest the operational efficiency of IS have positively impact on firms performance. Similarly, Greene and Segal (2004) show that efficiency increase profitability of the firms in terms of return on equity in the insurance industry. There is a strong positive link between the firm's value and the utilisation of the firm's resources.

4.6.2 Reliability (R)

The standardized beta coefficient is 0.118 which shows a 11.8% positive impact of the independent variable (Reliability) on dependent variable (Business Performance) with a significance value of 0.275 which is more than 0.05. This indicates that software reliability places an insignificant impact on the business performance of business firms in Malaysia.

This finding is similar to the study by Goodhue & Thompson (1995) that systems reliability had no impact on productivity and effectiveness and found no relationship between reliability and performance for individual users of ERP systems. On the other hand, this finding contradicts with Bharati & Chaudhury (2006) where they found a significant relationship in reliability of system to decision-making satisfaction in an e-commerce environment. Gorla et al., (2010) explores the linkage between IS quality and organizational impact . The results indicate that IS quality dimensions have a significant positive influence on organizational impact either directly or indirectly Accounting software system is reliable when information delivered on time and with error-free performance will result in timely and efficient decision making, which in turn leads to better internal organizational efficiency (Gorla et al., 2010).

4.6.3 Ease of Use (EOU)

The standardized beta coefficient is 0.365 which shows a 36.5% positive impact of the independent variable (Ease of Use) on dependent variable (Business Performance) with a significance value of 0.001 which is less than 0.05. This indicates that software ease of use places a significant impact on the business performance of business firms in Malaysia.

The finding of this study is supported by several other studies. Bias and Mayhew (2005) found an increase in ease of use positively influence several aspects of a company's output quality such as increased sales and revenues, productivity and customer satisfaction. Bharati and Chaudhury (2006) found a significant relationship between perceived ease of use and performance. Gefen and Straub (2000) explore the relative importance of perceived ease of use in IS adoption as to impact on firms performance. Findings by Goodhue & Thompson (1995) however, contradicted as they found systems perceived ease of use had no impact on productivity and effectiveness.

4.6.4 Data Quality (DQ)

The standardized beta coefficient is 0.087 which shows a 8.7% negative impact of the independent variable (Data Quality) on dependent variable (Business Performance) with a significance value of 0.329 which is more than 0.05. This indicates that software data quality places an insignificant impact on the business performance of business firms in Malaysia.

This finding contradicts with most of the trending empirical findings in literature. Few studies have scrutinized the effect of the AIS' data quality as critical success factor on organisational performance. Emeka-Nwokeji (2012) concluded that data quality is significant for the success of AIS as the quality of data is assured for enhancing firm's performance. Wixom & Watson (2001) assert data quality as one of the critical factor is relevant directly to effort for decision making. Al_Qudah and Shukeri (2014) noted data quality is strongly related and positively impacts the perception of company internal auditors. More scientific studies on found data quality and AIS performance are strongly related (Emeka-Nwokeji, 2012). The results from the study provide empirical support, suggesting that data quality positively influences the business performance. The analysis also suggests that data quality is an important antecedent of business performance.

4.6.5 Accuracy (A)

The standardized beta coefficient is 0.066 which shows a small 6.6% positive impact of the independent variable (Accuracy) on dependent variable (Business Performance) with a significance value of 0.462 which is more than 0.05. This indicates that software accuracy places an insignificant impact on the business performance of business firms in Malaysia.

According to this research, accuracy has no significant influential role on business performance in the context of this research. This finding however contradicts with past research findings. Swanson (1997) found non conformation of information provided by an IS leads to disruption of operations in the organization and hefty maintenance costs, ensuing in high costs to the organization. Banker et al. (1990) assert that firm will benefit from reduced labour costs and waste when the operational information from accounting software is highly accurate.

5.0 CONCLUSION & RECOMMENDATION

This research was conducted to investigate the impact accounting software characteristic in business performance of firms in Malaysia. The quantitative data required for the study was gathered through a sample size of 78 participants

(accountants or employees who involve in using accounting software in their work). Regression analysis was employed to gauge the impact of independent variables (Efficiency, Reliability, Ease of Use, Data Quality and Accuracy) on the Dependent Variable (Business Performance).

All the variables (dependent and independent) chosen for the study have indicated Cronbach's alpha scoring of above 0.80, indicating all variables have high internal consistency and are acceptable. The descriptive data analysis also supported that most of the respondents perceived that the independent variables chosen for this study affects positively to the dependent variable. The regression analysis indicated two variables (Efficiency and Ease of Use) out of the five independent variables have significant impact on business performance. Meanwhile, the other three variables Reliability, Data Quality and Accuracy were not found to have a significant impact on business performance.

In conclusion accounting software systems is of great importance and has a great value to businesses, organization and the economy. The accurate and reliable information flow is very crucial to the growth of economy. Performance management play a key role in improving the overall value of an organization. Prior researches have shown that accounting information system adoption does increases firm's performance, profitability, and efficiency operations. This study showed that there is strong relationship between the characteristic of accounting software and business performance, which means access to accurate accounting information, will lead to organizational effectiveness. Therefore, it can be concluded that accounting software has an impact on business performance of firms in Malaysia.

5.1 Research limitations

This study comprises of few limitations. Firstly, this study was conducted in English. The study could have included the survey questions in the local language, Bahasa Malaysia. This would make the survey sample more evenly distributed among the population and would eliminate misinterpretation of the survey questions. Secondly, this research was done generally on any business organization in Malaysia without specifically looking into one industry. Hence, different industry may have different requirement on the accounting software which may have different impact on the business performance. Another limitation is the difficulty in getting feedback on the questionnaire from respondents due to time constraints.

5.2 Future research directions

Throughout this research work, the suggestions concerning the expansion of the present study ascend. First, in data collection, I would suggest to collect data from disparate sources. Secondly, I would suggest similar studies to be done in different industries to compare the findings. Finally, a similar study could be carried out focusing on the effectiveness of accounting software systems in improving the business performance. Similarly, a study could also be carried out focusing on factors affecting the implementation of accounting software systems or challenges faced during implementation of accounting software systems in Malaysia.

References

- Al_Qudah, H. M. A. & Shukeri, S. N. B. (2014). The Role Of Data Quality And Internal Control In Raising The Effectiveness Of Ais In Jordan Companies. *International Journal of Technology Enhancements and Emerging Engineering Research*, 3(8), 298–303.
- Alford, A. and J. Boatsman (1995). Predicting long-term stock return volatility: Implications for accounting and valuation of equity derivatives. *The Accounting Review* (October): 599 - 618.
- Al-Hakim, L. (2007). Information quality management: theory and applications (1st ed.). U K: IGI Global.
- Ali, B.J., Omar, W.A.W. and Bakar, R. (2016). ACCOUNTING INFORMATION SYSTEM (AIS) AND ORGANIZATIONAL PERFORMANCE: MODERATING EFFECT OF ORGANIZATIONAL CULTURE.
- Banker, R.D., Kauffman, R.J. and Morey, R.C., 1990. Measuring gains in operational efficiency from information technology: a study of the Positran deployment at Hardee's Inc. *Journal of Management Information Systems*, 7(2), pp.29-54.
- Bias, R.G. and Mayhew, D.J. eds., 2005. *Cost-justifying usability: An update for the Internet age*. Elsevier.
- Brady, H.E. and Johnson, R., 2008. The rolling cross section and causal distribution. *Ann Arbor, MI: University of Michigan Press*.
- Brynjolfsson, E. and Hitt, L., 1996. Paradox lost? Firm-level evidence on the returns to information systems spending. *Management science*, 42(4), pp.541-558.
- Brynjolfsson, E. and Hitt, L.M., 2000. Beyond computation: Information technology, organizational transformation and business performance. *The Journal of Economic Perspectives*, 14(4), pp.23-48.
- Casler, D. and T. Hall. 1985. Firm-specific asset valuation accuracy using a composite price index. *Journal of Accounting Research* (Spring): 110-122.
- Cerny, C.A., & Kaiser, H.F. (1977). A study of a measure of sampling adequacy for factor-analytic correlation matrices. *Multivariate Behavioral Research*, 12(1), 43-47.
- Chan YE, Huff SL, Barclay DW, Copeland DG, 1997. Business strategic orientation, information systems strategic orientation and strategic alignment. *Inf Syst Res* 1997;8(2):125– 50.
- Chandar, N. and Bricker, R., 2002. Incentives, Discretion, and Asset Valuation in Closed-End Mutual Funds. *Journal of Accounting Research*, 40(4), pp.1037-1070.
- Choe. J M (2002) The Organizational Learning Effect of Management Accounting Information Under Advanced Manufacturing Technology, *European Journal of Information systems*, 11: 142-158.
- Clikeman, P.M., 1999. Improving information quality. *Internal Auditor*, 56(3), pp.32-34.
- Cragg PB, King M, Hussin H. (2002), IT alignment and firm performance in small manufacturing firms. *Journal of Strategic Information System*; 11:109–32.
- Creswell, J. W. (2009). *Research design: Qualitative and quantitative approaches* (3rd ed.). London: SAGE Publication.
- Cronbach, L. J., 1951. Coefficient alpha and the internal structure of tests. *Psychometrika*. 16(3), Pp. 297-334.
- DeVellis, R.F. (2012). *Scale development: Theory and applications*. Los Angeles: Sage. pp. 109–110.

- Ismail, N.A. and King, M., 2005. Firm performance and AIS alignment in Malaysian SMEs. *International Journal of Accounting Information Systems*, 6(4), pp.241-259.
- Ismail, N.A. and King, M., 2014. Factors influencing the alignment of accounting information systems in small and medium sized Malaysian manufacturing firms. *Journal of Information Systems and Small Business*, 1(1-2), pp.1-20.
- Judd, C. M., & McClelland, G. H. (1989). *Data analysis: A model comparison approach*. New York: Harcourt Brace Jovanovich.
- Kaiser, H. F. (1960). The application of electronic computers to factor analysis. *Educational and Psychological Measurement*, 20(1), 141-151. doi:10.1177/001316446002000116.
- Kouser, R., Awan, A., Rana, G. and Shahzad, F., 2011. Firm Size, Leverage and Profitability: Overriding Impact of Accounting Information System. *Journal of Management and Business Review*, 1(10), pp.58-64.
- Landauer, T. K. (1996). *The trouble with computers*. Cambridge, MA, The MIT Press. p158.
- Lee, C.S., 2001. Modeling the business value of information technology. *Information & Management*, 39(3), pp.191-210.
- Levy, M. Powel, P & Yetton, P. (2011). Contingent dynamics of IS strategic alignment in small & medium sized enterprises. *Journal of Systems & Information Technology*, 13(2): 106-124.
- Maines, L.A. and Wahlen, J.M., 2006. The nature of accounting information reliability: Inferences from archival and experimental research. *Accounting Horizons*, 20(4), pp.399-425.
- Osei-Bryson, K.M. and Ko, M., 2004. Exploring the relationship between information technology investments and firm performance using regression splines analysis. *Information & management*, 42(1), pp.1-13.
- Pérez Estébanez, R., Urquía Grande, E. and Muñoz Colomina, C., 2010. Information technology implementation: evidence in Spanish SMEs. *International Journal of Accounting & Information Management*, 18(1), pp.39-57.
- Phillips, D. C., & Burbules, N. C. (2000). *Postpositivism and educational research*. Lanham, MD: Rowman & Littlefield.
- Ravichandran, T. and Rai, A., 2000. Quality management in systems development: an organizational system perspective. *MIS quarterly*, pp.381-415.
- Raymond L, Pare G, Bergeron F. Matching information technology and organisational structure: An empirical study with implications for performance. *Eur J Inf Syst* 1995;4:3- 16.
- Rogers, P. R., Miller, A., & Judge, W. Q. (1999). Using information-processing theory to understand planning/performance relationships in the context of strategy. *Strategic Management Journal*, 20, 567-577.
- Saira, K., Zariyawati, M.A. and Annuar, M.N., 2010. Information system and firms' performance: the case of Malaysian small medium enterprises. *International business research*, 3(4), p.28.
- Sajady H, Dastgir M, Nejad HH (2008) Evaluation of the effectiveness of accounting information systems. *International Journal of Information Science and Management* 6: 49-59.
- Sam, M., Fazli, M., Hoshino, Y. and Tahir, M.N.H., 2012. The adoption of computerized accounting system in small medium enterprises in Melaka, Malaysia.

- Sharma, B. and Gadenne, D., 2002. An inter-industry comparison of quality management practices and performance. *Managing Service Quality: An International Journal*, 12(6), pp.394-404.
- Sharma, M. & Bhagwat, R. (2003). Performance measurements in the implementation of information systems in small and medium sized enterprises: A framework and empirical analysis. *Measuring Business Excellence*, 10(4): 8-21.
- Shriver, K. 1986. Further evidence on the marginal gains in accuracy of alternative levels of specificity of the producer price indexes. *Journal of Accounting Research* Vol. 24, No. 1 (Spring): 151-165.
- Shriver, K. 1987. An empirical examination of the potential measurement error in current cost data. *The Accounting Review* Vol. 62, No. 1 (January): 79-96.
- Sunder, S. and Waymire, G., 1983. Marginal gains in accuracy of valuation from increasingly specific price indexes: Empirical evidence for the US economy. *Journal of Accounting Research*, pp.565-580.
- Swanson, L., 1997. Computerized maintenance management systems: a study of system design and use. *Production and inventory management journal*, 38(2), p.11.
- Wickramsainghe, D.M.J., Pamarathna, R.M.M.D., Cooray, N.H.K. and Dissanayake, T.D.S.H., 2017. Impact of accounting software for Business Performance. *Imperial Journal of Interdisciplinary Research*, 3(5).
- Wilkinson JW, Cerullo MJ, Raval V, Wong-On-Wing B (2000) *Accounting information systems: Essential concepts and applications*. John Wiley and Sons, New York.
- Wixom, B. H. & Watson, H. J. (2001). An empirical investigation of the factors affecting data warehousing success. *MIS quarterly*, 25(1), 17-41. JSTOR.
- Xu, H. (2003). Critical success factors for accounting information systems data quality.
- Xu, H. (2009). Data quality issues for accounting information systems' implementation: Systems, stakeholders, and organizational factors. *Journal of Technology Research*, 1, 1. Academic and Business Research Institute (AABRI).
- Zikmund, W.G., Babin, B.J. and Carr, J.C., dan Griffin, M.(2010). *Business research methods*.
- Zygmunt, C. & Smith, M. R. (2014). Robust factor analysis in the presence of normality violations, missing data, and outliers: Empirical questions and possible solutions. *The Quantitative Methods for Psychology*, 10 (1), 40-55.